BTS-III-11.18-1713 Reg. No. [ ]

B

B.Tech. Degree III Semester Examination November 2018

ME 15-1302 ELECTRICAL TECHNOLOGY
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

($10 \times 2 = 20$)

I. (a) Why does the field winding of a DC series machine have less number of turns than that of a DC shunt machine?
(b) DC shunt motor is considered as a constant speed motor. Why?
(c) Explain the speed-armature current characteristic of DC series and shunt motor.
(d) A 50 kVA single phase transformer has a turns ratio of 300/20. The primary winding is connected to 2200 V, 50 Hz supply. Calculate:
  (i) The secondary voltage at no load.
  (ii) The approximate values of the primary and secondary currents on full load.
  (iii) The maximum value of the flux.
(e) What are the different losses in transformer? Derive the condition for maximum efficiency of a transformer.
(f) Why OC test in a transformer is normally carried on low voltage side?
(g) Derive an expression for the winding factor of an alternator.
(h) Why are damper windings used in synchronous motors?
(i) What is skin effect?
(j) Draw the schematic diagram of a nuclear power plant.

PART B

($4 \times 10 = 40$)

II. (a) Explain the function of commutator in a DC machine. What are the effects of poor commutation on the performance of a DC machine? What methods are used to overcome these difficulties? (6)
(b) A 4 pole, 460 V shunt motor has its armature wave wound with 888 conductors. The useful flux per pole is 0.02 wb and the resistance of the armature circuit is 0.7Ω. If the armature current is 40 A, calculate: (4)
  (i) The speed.
  (ii) The torque in Nm.

OR

III. (a) A 230 V shunt motor, running on no load and at normal speed, takes an armature current of 2.5 A from 230 V mains. The field circuit resistance is 230Ω and the armature circuit resistance is 0.3Ω. Calculate the motor output, in kW, and the efficiency when the total current taken from the mains is 35 A. If the motor is used as a 230 V shunt generator, find the efficiency and input power for an output current of 35 A. (6)
(b) Discuss the methods of speed control of a DC series motor. (4)

(P.T.O.)

IV. A single phase transformer is rated at 10 kVA, 230/100 V. When the secondary terminals are open circuited and the primary winding is supplied at normal voltage (230 V), the current input is 2.6 A at a power factor of 0.3. When the secondary is short circuited, a voltage of 18 V applied to the primary causes the full load current (100 A) to flow in the secondary, the power input to the primary being 240 W. Draw the equivalent circuit referred to LV side. Calculate: (10)

(i) The efficiency of the transformer at full load, unity power factor.
(ii) The load at which maximum efficiency occur.
(iii) The value of the maximum efficiency.

OR

V. (a) Draw and explain the vector diagram of a practical transformer when it is connected to a capacitive load. (5)

(b) A 2000/200 V, 20 kVA transformer is connected as a step up transformer in which input in connected to the 2000 V winding. Calculate: (5)

(i) the LV and HV side voltage ratings of auto transformer.
(ii) its kVA rating.
(iii) kVA transferred inductively and conductively.
(iv) its efficiency at full load 0.8 power factor lagging.

VI. (a) Explain the effect of varying excitation on armature current and power factor in a synchronous motor. Draw V curves. (4)

(b) The data obtained on 100 kVA, 1100 V, 3 phase alternator is: (6)
Open circuit test: Field current = 12.5 A dc, line voltage = 420 V ac
Short circuit test: Field current = 12.5 A dc, line current = rated value.
The ac resistance of the winding per phase is 0.5Ω. Calculate the voltage regulation of alternator at 0.8 pf lagging and 0.8 pf leading.

OR

VII. An 8 pole, 50 Hz, 3ϕ induction motor is running at a speed of 710 rpm with an input power of 35 kW. The stator copper loss at this condition is known to be 1200 W while the rotational losses are 600 W. Find: (10)

(i) the rotor copper loss.
(ii) the gross torque developed.
(iii) the gross mechanical power developed.
(iv) the net torque and mechanical power output.

VIII. (a) What are the advantages of high voltage transmission? State its limitations. (4)

(b) Discuss any five electrical equipments in substation. (6)

OR

IX. (a) Define corona. What are the factors affecting corona? (4)

(b) Explain hydro electric power generation with neat sketch. (6)

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